



GSFA Online

Updated up to the 48th Session of the Codex Alimentarius Commission (2025)

FOOD CATEGORY DETAILS

Frozen fish, fish fillets, and fish products, including mollusks, crustaceans, and echinoderms (09.2.1)

Description:

Fresh, including partially cooked, fish subjected to freezing or quick-freezing at sea and on land for further processing.¹ Examples include: frozen or deep frozen clams, cod fillets, crab, finfish, haddock, hake, lobster, minced fish, prawns and shrimp; frozen fish roe; frozen surimi; and frozen whale meat.

This page provides information on the food additive provisions that are acceptable for use in foods conforming to the food category.

This food category is listed in the **Annex to Table 3**. Unless specifically indicated below, food additive provisions implied by **Table 3** do not automatically apply to this category.

GSFA Provisions for Food Category 09.2.1

INS No.	Food Additive or Group	Max Level	Notes	Defined In
586	4-Hexylresorcinol	50 mg/kg	 Note XS36  Note XS312  Note 683  Note XS315  Note 641  Note XS95  Note XS165  Note XS191  Note XS190  Note XS92  Note XS292	09.2.1
	ASCORBYL ESTERS	1,000 mg/kg	 Note XS36  Note XS312  Note 10  Note XS315  Note XS95  Note XS191  Note XS190  Note XS92  Note 392	09.2.1

INS No. ▼	Food Additive or Group	Max Level ▼	Notes ▼	Defined In ▼
950	Acesulfame potassium	200 mg/kg	Note XS236 Note 188 Note XS36 Note XS312 Note XS191 Note XS190 Note XS311 Note XS92 Note 144 Note XS189 Note XS166 Note XS292 Note XS244 Note XS315 Note XS95 Note XS167 Note 478 Note XS165 Note XS222	09.2
472a	Acetic and fatty acid esters of glycerol	GMP	Note XS36 Note XS312 Note XS315 Note XS95 Note XS165 Note XS191 Note XS190 Note XS292 Note XS92	09.2.1
1414	Acetylated distarch phosphate	GMP	Note XS36 Note XS312 Note XS315 Note XS95 Note XS165 Note XS191 Note XS190 Note XS292 Note XS92	09.2.1
969	Advantame	3 mg/kg	Note 144	09.2
406	Agar	GMP	Note XS36 Note XS312 Note 53 Note XS315 Note XS95 Note XS165 Note XS191 Note XS190 Note 3 Note XS92 Note XS292	09.2.1
400	Alginic acid	GMP	Note XS36 Note XS312 Note XS315 Note XS95 Note XS165 Note XS191 Note XS190 Note XS92 Note 16 Note XS292	09.2.1
129	Allura red AC	300 mg/kg	Note XS36 Note XS312 Note XS315 Note XS95 Note XS165 Note XS191 Note XS190	09.2.1

INS No. ▼	Food Additive or Group	Max Level ▼	Notes ▼	Defined In ▼
			☐ Note XS92 ☐ Note XS292 ☐ Note 95	
403	Ammonium alginate	GMP	☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note XS292 ☐ Note XS92	09.2.1
300	Ascorbic acid, L-	GMP	☐ Note XS236 ☐ Note XS222 ☐ Note XS312 ☐ Note XS315 ☐ Note XS189 ☐ Note XS191 ☐ Note XS190 ☐ Note 307 ☐ Note 392	09.2.1
951	Aspartame	300 mg/kg	☐ Note XS236 ☐ Note XS36 ☐ Note XS312 ☐ Note 191 ☐ Note XS191 ☐ Note XS190 ☐ Note XS311 ☐ Note XS92 ☐ Note 144 ☐ Note XS189 ☐ Note XS166 ☐ Note XS292 ☐ Note XS244 ☐ Note XS315 ☐ Note XS95 ☐ Note XS167 ☐ Note 478 ☐ Note XS165 ☐ Note XS222	09.2
962	Aspartame-acesulfame salt	200 mg/kg	☐ Note XS236 ☐ Note XS244 ☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS167 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note XS311 ☐ Note 113 ☐ Note XS92 ☐ Note XS222 ☐ Note 144 ☐ Note XS189 ☐ Note XS166 ☐ Note XS292	09.2
133	Brilliant blue FCF	500 mg/kg	☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note XS92 ☐ Note XS292	09.2.1

INS No.	Food Additive or Group	Max Level	Notes	Defined In
320	Butylated hydroxyanisole (BHA)	200 mg/kg	 Note 95	09.2.1
			 Note XS36	
			 Note XS312	
			 Note XS315	
			 Note XS95	
			 Note XS165	
			 Note XS191	
			 Note XS190	
			 Note 15	
			 Note XS92	
			 Note 392	
 Note 180				
321	Butylated hydroxytoluene (BHT)	200 mg/kg	 Note XS36	09.2.1
			 Note XS312	
			 Note XS315	
			 Note XS95	
			 Note XS165	
			 Note XS191	
			 Note XS190	
			 Note 15	
			 Note XS92	
			 Note 392	
			 Note 180	
CAROTENES, BETA-		100 mg/kg	 Note XS236	09.2
			 Note XS244	
			 Note XS36	
			 Note XS312	
			 Note XS315	
			 Note XS95	
			 Note XS167	
			 Note XS165	
			 Note XS191	
			 Note XS190	
			 Note XS311	
			 Note XS92	
			 Note XS222	
			 Note 344	
			 Note XS189	
 Note 341				
 Note XS292				
404	Calcium alginate	GMP	 Note XS36	09.2.1
			 Note XS312	
			 Note XS315	
			 Note XS95	
			 Note XS165	
			 Note XS191	
			 Note XS190	
			 Note XS292	
			 Note XS92	
302	Calcium ascorbate	GMP	 Note 308	09.2.1
			 Note XS36	
			 Note XS312	
			 Note XS315	
			 Note XS95	
			 Note XS165	
			 Note XS191	
			 Note XS190	
			 Note XS92	
 Note 392				
170(i)	Calcium carbonate	GMP	 Note XS36	09.2.1
			 Note XS312	
			 Note XS315	
			 Note XS95	
			 Note XS165	
			 Note XS191	
			 Note XS190	
			 Note XS92	

INS No. ▼	Food Additive or Group	Max Level ▼	Notes ▼	Defined In ▼
			☐ Note XS292 ☐ Note 95	
161g	Canthaxanthin	35 mg/kg	☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note XS92 ☐ Note XS292 ☐ Note 95	09.2.1
150c	Caramel III - ammonia caramel	30,000 mg/kg	☐ Note XS236 ☐ Note XS244 ☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS167 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note XS311 ☐ Note XS92 ☐ Note XS222 ☐ Note XS189 ☐ Note XS166 ☐ Note XS292	09.2
150d	Caramel IV - sulfite ammonia caramel	30,000 mg/kg	☐ Note XS236 ☐ Note XS244 ☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS167 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note XS311 ☐ Note XS92 ☐ Note XS222 ☐ Note XS189 ☐ Note XS166 ☐ Note XS292 ☐ Note 95	09.2
120	Carmines	100 mg/kg	☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note XS92 ☐ Note XS292 ☐ Note 95 ☐ Note 178	09.2.1
410	Carob bean gum	GMP	☐ Note 391 ☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS191 ☐ Note XS190 ☐ Note XS292 ☐ Note XS92	09.2.1
160e	Carotenal, beta-apo-8'-	100 mg/kg	☐ Note XS236 ☐ Note XS244	09.2

INS No. ▼	Food Additive or Group	Max Level ▼	Notes ▼	Defined In ▼
			Note XS36 Note XS312 Note XS315 Note 439 Note XS95 Note XS167 Note XS165 Note XS191 Note XS190 Note XS311 Note XS92 Note 336 Note XS222 Note XS189 Note XS292	
160a(ii)	beta-Carotenes, vegetable	100 mg/kg	Note XS236 Note XS36 Note XS312 Note XS191 Note XS190 Note XS311 Note XS92 Note XS189 Note 341 Note XS292 Note XS244 Note XS315 Note 439 Note XS95 Note XS167 Note XS165 Note 336 Note XS222 Note 344	09.2
407	Carrageenan	GMP	Note 391 Note 332 Note XS36 Note XS312 Note XS315 Note XS95 Note XS191 Note XS190 Note XS92 Note XS292	09.2.1
330	Citric acid	GMP	Note 331 Note 391 Note XS36 Note XS312 Note XS315 Note XS95 Note XS191 Note XS190 Note 392	09.2.1
472c	Citric and fatty acid esters of glycerol	GMP	Note XS36 Note XS312 Note XS315 Note XS95 Note XS165 Note XS191 Note XS190 Note XS92 Note 392	09.2.1
1400	Dextrins, roasted starch	GMP	Note XS36 Note XS312 Note 53 Note XS315 Note XS95	09.2.1

INS No. ▼	Food Additive or Group	Max Level ▼	Notes ▼	Defined In ▼
			☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note 3 ☐ Note XS92 ☐ Note XS292	
627	Disodium 5'-guanylate		GMP ☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note XS92 ☐ Note XS292 ☐ Note 95	09.2.1
631	Disodium 5'-inosinate		GMP ☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note XS92 ☐ Note XS292 ☐ Note 95	09.2.1
635	Disodium 5'-ribonucleotides		GMP ☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note XS92 ☐ Note XS292 ☐ Note 95	09.2.1
	ETHYLENE DIAMINE TETRA ACETATES	75 mg/kg	☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note XS92 ☐ Note 392 ☐ Note 21	09.2.1
315	Erythorbic Acid (Isoascorbic acid)		GMP ☐ Note 308 ☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note XS92 ☐ Note 392	09.2.1
418(i)	Gellan gum		GMP ☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note XS292 ☐ Note XS92	09.2.1
412	Guar gum		GMP ☐ Note 391	09.2.1

INS No. ▼	Food Additive or Group	Max Level ▼	Notes ▼	Defined In ▼
			☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note 73 ☐ Note XS191 ☐ Note XS190 ☐ Note XS92 ☐ Note XS292	
414	Gum arabic (Acacia gum)	GMP	☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note XS92 ☐ Note 16 ☐ Note XS292	09.2.1
463	Hydroxypropyl cellulose	GMP	☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note XS92 ☐ Note 16 ☐ Note XS292	09.2.1
464	Hydroxypropyl methyl cellulose	GMP	☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note XS92 ☐ Note 16 ☐ Note XS292	09.2.1
1440	Hydroxypropyl starch	GMP	☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note XS292 ☐ Note XS92	09.2.1
132	Indigotine (Indigo carmine)	300 mg/kg	☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note XS92 ☐ Note XS292 ☐ Note 95	09.2.1

INS No. ▼	Food Additive or Group	▼ Max Level	▼ Notes	▼ Defined In ▼
416	Karaya gum		GMP  Note XS36  Note XS312  Note XS315  Note XS95  Note XS165  Note XS191  Note XS190  Note XS292  Note XS92	09.2.1
425	Konjac flour		GMP  Note XS36  Note XS312  Note XS315  Note XS95  Note XS165  Note XS191  Note XS190  Note XS92  Note 16  Note XS292	09.2.1
472b	Lactic and fatty acid esters of glycerol		GMP  Note XS36  Note XS312  Note XS315  Note XS95  Note XS165  Note XS191  Note XS190  Note XS292  Note XS92	09.2.1
322(i)	Lecithin		GMP  Note XS36  Note XS312  Note XS315  Note XS95  Note XS165  Note XS191  Note XS190  Note XS92  Note 392	09.2.1
511	Magnesium chloride		GMP  Note XS36  Note XS312  Note XS315  Note XS95  Note XS165  Note XS191  Note XS190  Note XS292  Note XS92	09.2.1
421	Mannitol		GMP  Note XS36  Note XS312  Note XS315  Note XS95  Note XS165  Note XS191  Note XS190  Note XS292  Note XS92	09.2.1
461	Methyl cellulose		GMP  Note 391  Note 332  Note XS36  Note XS312  Note XS315  Note XS95  Note XS191  Note XS190  Note XS92  Note XS292	09.2.1
465	Methyl ethyl cellulose		GMP  Note XS36	09.2.1

INS No. ▼	Food Additive or Group	Max Level ▼	Notes ▼	Defined In ▼
			☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note XS292 ☐ Note XS92	
460(i)	Microcrystalline cellulose (Cellulose gel)	GMP	☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note XS92 ☐ Note 16 ☐ Note XS292	09.2.1
621	Monosodium L-glutamate	GMP	☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note XS92 ☐ Note XS292 ☐ Note 95	09.2.1
942	Nitrous oxide	GMP	☐ Note 308 ☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note XS92 ☐ Note 392	09.2.1
1404	Oxidized starch	GMP	☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note XS292 ☐ Note XS92	09.2.1
	PHOSPHATES	2,200 mg/kg	☐ Note 394 ☐ Note XS36 ☐ Note 393 ☐ Note XS312 ☐ Note 33 ☐ Note XS191 ☐ Note XS292	09.2.1
440	Pectins	GMP	☐ Note 391 ☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS191 ☐ Note XS190 ☐ Note XS92 ☐ Note 16 ☐ Note XS292	09.2.1
1200	Polydextroses	GMP	☐ Note XS36 ☐ Note XS312	09.2.1

INS No. ▼	Food Additive or Group	Max Level ▼	Notes ▼	Defined In ▼
			Note XS315 Note XS95 Note XS165 Note XS191 Note XS190 Note XS292 Note XS92	
475	Polyglycerol esters of fatty acids	5,000 mg/kg	Note 241	09.2.1
124	Ponceau 4R (Cochineal red A)	30 mg/kg	Note XS36 Note XS312 Note XS315 Note XS95 Note XS165 Note XS191 Note 395 Note XS190 Note XS292	09.2.1
402	Potassium alginate	GMP	Note XS36 Note XS312 Note XS315 Note XS95 Note XS165 Note XS191 Note XS190 Note XS292 Note XS92	09.2.1
508	Potassium chloride	GMP	Note XS36 Note XS312 Note XS315 Note XS95 Note XS165 Note XS191 Note XS190 Note XS292 Note XS92	09.2.1
332(i)	Potassium dihydrogen citrate	GMP	Note XS236 Note 391 Note XS244 Note XS36 Note XS312 Note XS315 Note XS95 Note XS167 Note XS191 Note XS190 Note XS311 Note XS92 Note 253 Note XS222 Note XS189 Note XS292	09.2
460(ii)	Powdered cellulose	GMP	Note XS36 Note XS312 Note XS315 Note XS95 Note XS165 Note XS191 Note XS190 Note XS92 Note 16 Note XS292	09.2.1
407a	Processed eucheuma seaweed (PES)	GMP	Note 391 Note 332 Note XS36 Note XS312 Note XS315	09.2.1

INS No. ▼	Food Additive or Group	Max Level ▼	Notes ▼	Defined In ▼
			Note XS95 Note XS191 Note XS190 Note XS92 Note XS292	
	RIBOFLAVINS		GMP Note XS36 Note XS312 Note XS315 Note XS95 Note XS165 Note XS191 Note XS190 Note XS92 Note XS292 Note 95	09.2.1
	SULFITES	100 mg/kg	Note XS36 Note XS312 Note XS315 Note 139 Note XS165 Note XS191 Note XS190 Note 19 Note 392 Note 44	09.2.1
470(i)	Salts of myristic, palmitic and stearic acids with ammonia, calcium, potassium and sodium		GMP Note XS36 Note XS312 Note XS315 Note XS95 Note XS165 Note XS191 Note 71 Note XS190 Note XS92 Note XS292	09.2.1
470(ii)	Salts of oleic acid with calcium, potassium and sodium		GMP Note XS36 Note XS312 Note XS315 Note XS95 Note XS165 Note XS191 Note XS292 Note XS190 Note XS92	09.2.1
401	Sodium alginate		GMP Note XS36 Note XS312 Note XS315 Note XS95 Note XS191 Note XS292 Note XS92	09.2.1
301	Sodium ascorbate		GMP Note XS236 Note XS244 Note XS312 Note XS315 Note XS167 Note XS191 Note 307 Note 437 Note XS92 Note 392 Note XS222 Note XS189	09.2
466	Sodium carboxymethyl cellulose (Cellulose gum)		GMP Note 391 Note 332 Note XS36	09.2.1

INS No. ▼	Food Additive or Group	Max Level ▼	Notes ▼	Defined In ▼
			☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS191 ☐ Note XS190 ☐ Note XS92 ☐ Note XS292	
331(i)	Sodium dihydrogen citrate	GMP	☐ Note XS236 ☐ Note 391 ☐ Note XS244 ☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS167 ☐ Note XS191 ☐ Note XS190 ☐ Note XS311 ☐ Note XS92 ☐ Note 253 ☐ Note XS222 ☐ Note XS189 ☐ Note XS292	09.2
316	Sodium erythorbate (Sodium isoascorbate)	GMP	☐ Note 308 ☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note XS92 ☐ Note 392	09.2.1
576	Sodium gluconate	GMP	☐ Note XS236 ☐ Note XS244 ☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS167 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note XS311 ☐ Note XS92 ☐ Note XS222 ☐ Note XS189 ☐ Note XS166 ☐ Note XS292	09.2
110	Sunset yellow FCF	300 mg/kg	☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190 ☐ Note XS92 ☐ Note XS292 ☐ Note 95	09.2.1
417	Tara gum	GMP	☐ Note XS36 ☐ Note XS312 ☐ Note XS315 ☐ Note XS95 ☐ Note 73 ☐ Note XS165 ☐ Note XS191 ☐ Note XS190	09.2.1

INS No. ▼	Food Additive or Group	Max Level ▼	Notes ▼	Defined In ▼
413	Tragacanth gum	GMP	Note XS92	09.2.1
			Note XS292	
			Note XS36	
			Note XS312	
			Note XS315	
			Note XS95	
			Note XS165	
			Note XS191	
			Note XS190	
			Note XS292	
333(iii)	Tricalcium citrate	GMP	Note XS92	09.2
			Note XS236	
			Note XS244	
			Note XS36	
			Note XS312	
			Note XS315	
			Note XS95	
			Note XS167	
			Note XS165	
			Note XS191	
			Note XS190	
			Note XS311	
			Note XS92	
			Note XS222	
332(ii)	Tripotassium citrate	GMP	Note XS189	09.2
			Note XS166	
			Note XS292	
			Note XS236	
			Note 391	
			Note XS244	
			Note XS36	
			Note XS312	
			Note XS315	
			Note XS95	
			Note XS167	
			Note XS191	
			Note XS190	
			Note XS311	
331(iii)	Trisodium citrate	GMP	Note XS92	09.2
			Note 253	
			Note XS222	
			Note XS189	
			Note XS292	
			Note XS236	
			Note 391	
			Note XS244	
			Note XS36	
			Note XS312	
			Note XS315	
			Note XS95	
			Note XS167	
			Note XS191	
415	Xanthan gum	GMP	Note XS190	09.2.1
			Note XS311	
			Note XS92	
			Note XS222	
			Note 253	
			Note XS165	
			Note XS191	
			Note XS95	
			Note XS315	
			Note XS312	
			Note XS36	
			Note XS292	

INS No. ▼	Food Additive or Group ▼	Max Level ▼	Notes ▼	Defined In ▼
Note XS92				
Items in uppercase (e.g. PHOSPHATES) refer to food additive groups. 1. Food Chemistry, H.-D. Belitz & W. Grosch, Springer-Verlag, Heidelberg, 1987, pp. 464-68.  printer-friendly version GSFA Home ◯ Food Categories ◯ Food Additives ◯ Search ◯ Functional Classes ◯ Glossary © FAO and WHO 2026				